

Hide menu

Voronoi Diagrams

Voronoi Diagrams: Structure

Complexity of Voronoi Diagrams

Video: Complexity of Voronoi Diagrams4 min

Practice Quiz: VoronoiStarted

Delanunay Triangulations

Angle-Optimal Triangulations

Legal Triangulations

Randomized Incremental Construction

Randomized Incremental Construction: Analysis

Programming: Voronoi Diagrams and Delaunay Triangulations

Wrap-Up Quiz

🎉 Congratulations! You passed!

Grade received 100% To pass 65% or higher

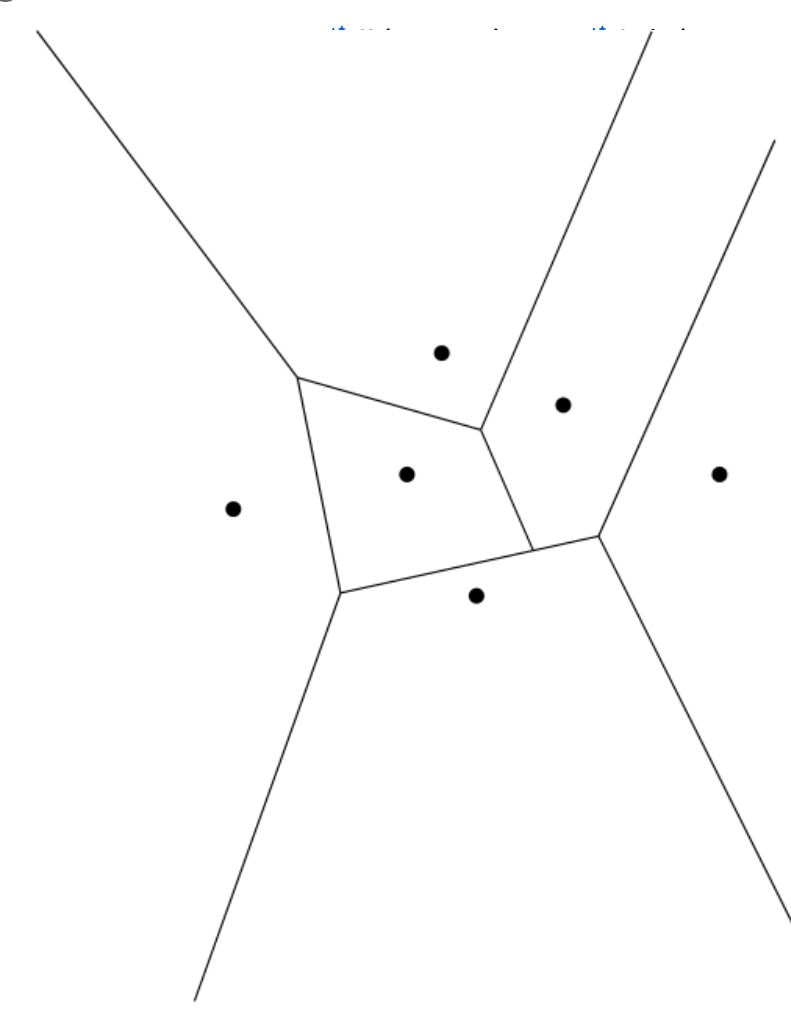
Voronoi

Go to next item

1. Which of the following diagrams is a correct Voronoi diagram?

1 / 1 point

Ⓐ:

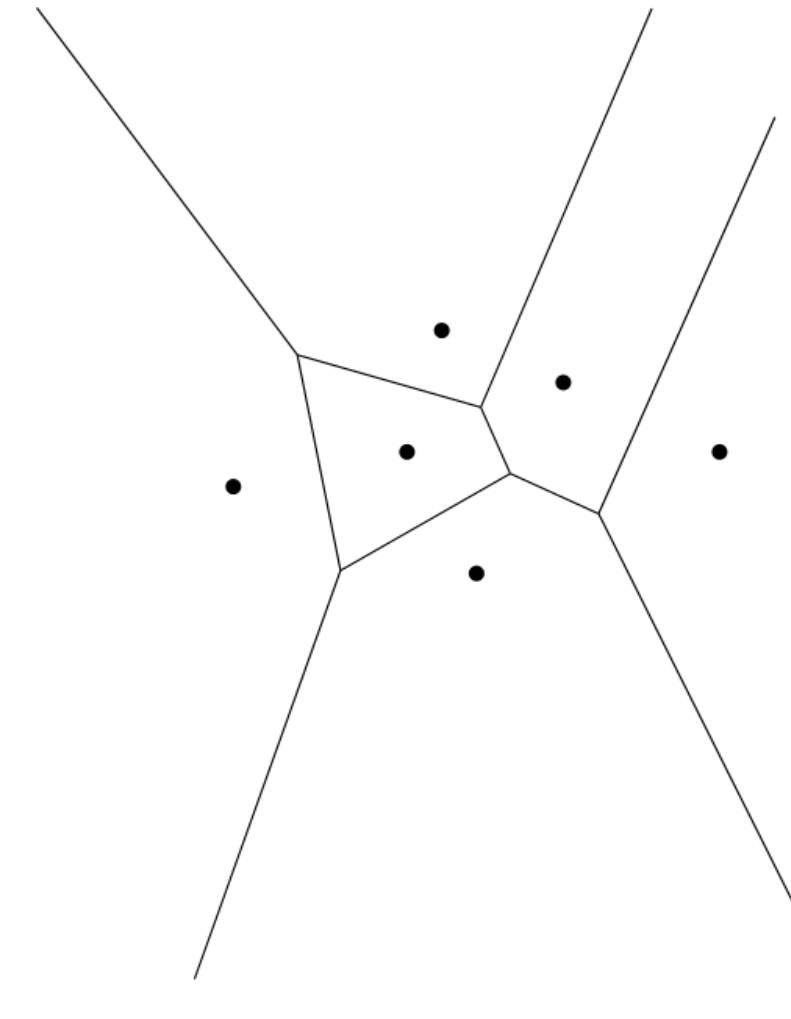


Try again

Your grade100%

View FeedbackWe keep your highest score

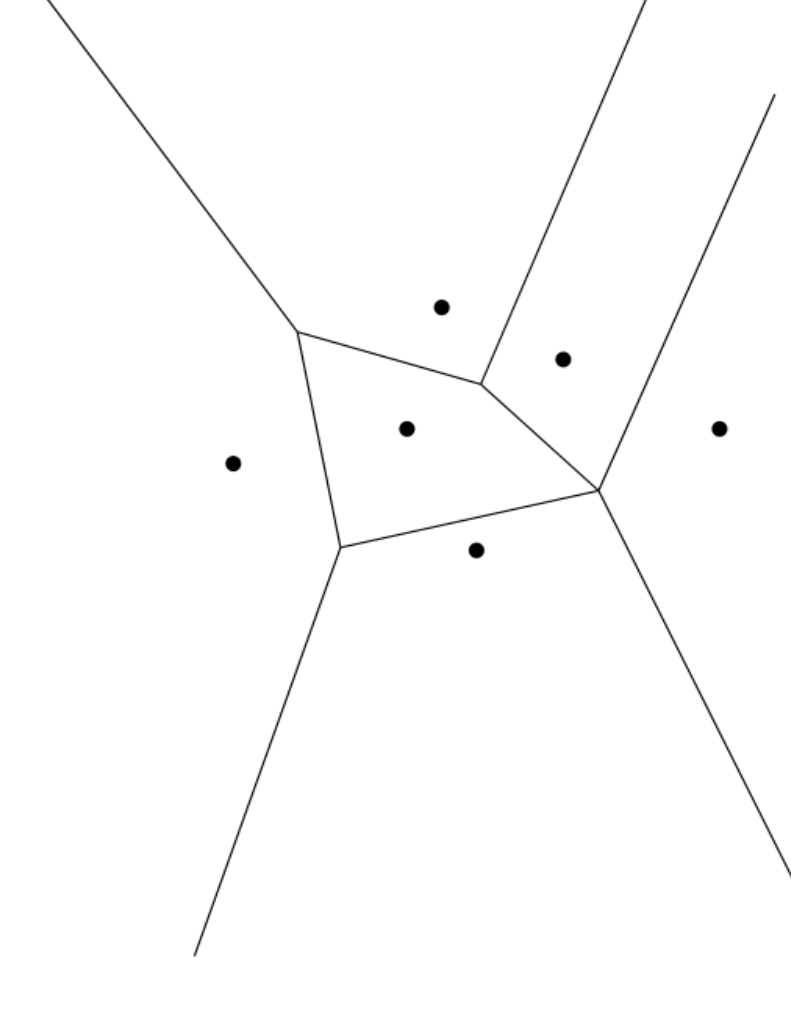
Ⓑ:



Correct

Correct, every point inside a cell is closest to the vertex of that cell.

Ⓒ:



Correct

Correct, for each component i we have |V_i| - |E_i| + |F_i| = 2, but when summing over all components we should not count the outermost face k times but only once.

2. Consider a connected plane graph G with 17 vertices and 45 edges. How many faces does G have (including the outer face)?

1 / 1 point

30

Correct

Correct! V - E + F = 2

3. Consider Euler's formula again for connected graphs. We want to derive a similar result for graphs with k connected components instead of connected graphs. What would this result be?

1 / 1 point

Ⓐ

$|V| - |E| + |F| = 2 \cdot k$

Ⓑ

$|V| - |E| + |F| = k + 1$

Ⓒ

$|V| - |E| + |F| = 2 \cdot k + 1$

Ⓓ

$|V| - |E| + |F| = 2 \cdot k - 1$

Correct

Correct, for each component i we have $|V_i| - |E_i| + |F_i| = 2$, but when summing over all components we should not count the outermost face k times but only once.

https://www.coursea.org/learn/geometric-algorithms/quiz/BPAKtIvoronoiattempt?redirectToCover=true

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